

## Final PSA figure plan

Direct execution authorized by user: replace purge with 5 kPa vacuum and include crystal volume, mass-transfer and heat coupling. Six panels: 298 K reference isotherms, total/partial pressure, loadings, temperature, solid volume, effective LDF coefficient. Separate rate plot and conceptual flowsheet. Read numerical CSV without smoothing; label all curves simulations; export PNG/SVG/PDF.

Method: user-specified anchors and gate width; local component/energy inventory equations; SciPy BDF documentation inspected at [https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve\\_ivp.html](https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html). 40 kJ/mol interpreted as effective heat per mol CO<sub>2</sub>, no duplicate phase heat. Unknown thermal gate relation explicitly compared with a fixed-threshold case.